SOC_MC Experimental Results: Loss-Landscape Theory of Mode Collapse

1. Experimental Setup

1.1 Overview

This document reports results from a five-goal experimental program designed to verify the loss-landscape theory of mode collapse in category (a) Stochastic Optimal Control (SOC) samplers. The theory predicts that mode-collapsed controllers are local minima of the training loss, and that the stability of a collapsed state depends on its loss value relative to a theoretical threshold.

1.2 Samplers Tested

Two category (a) SOC samplers, both ported from the Stein_ASBS codebase:

Sampler	Architecture	Controller dim	Loss
Adjoint Sampling (AS)	FourierMLP, boundary-only VE-SDE	83,330 params	Adjoint matching L2
ASBS	FourierMLP + corrector network, IPF	166,660 params	Bridge matching L2

Both use VE-SDE with sigma_min=0.01, sigma_max=3.0, giving g_max=3.0. Controller network: FourierMLP with sinusoidal time embeddings, SiLU activations.

1.3 Benchmarks

Three 2D Gaussian mixture benchmarks with K=5 modes at regular pentagon vertices (R=8.0, sigma_mode=0.8):

Benchmark	Hardness Axis	Weights	Covariances	Mode Positions
W5	Weight imbalance	Geometric decay r=0.5: (0.52, 0.26,	All isotropic, sigma=0.8	Pentagon, R=8.0
C5	Covariance heterogeneity	Equal (0.2 each)	Modes 0-2: sigma=0.8; Mode 3: sigma	Pentagon, R=8.0
B5	Barrier heterogeneity	Equal (0.2 each)	All isotropic, sigma=0.8	Modes 0-2 clustered at (-3,0),(0,0)

Adjacent-mode distance on pentagon: ~9.4, giving rho_sep ~4.7. B5 cluster separation: 3.0 (within), 12.0 (to isolated).

1.4 Subset Coverage

All 30 proper non-trivial subsets per benchmark tested exhaustively:

S	**=1: 5 subsets,**	**S**	**=2: 10 subsets,**	**S**	**=3: 10 subsets,**	**S**

Predicted easy subsets: subsets of {0,1,2} (high-weight / wide / clustered modes).

Predicted hard subsets: subsets containing mode 3 or 4.

1.5 Training Protocol

Parameter	Pre-train (Stage 1)	Stability (Stage 2)
Epochs	500	200
Target	Restricted p_S	Full p
Seeds	1 (seed 0)	5 (seeds 0-4)
Optimizer	Adam, lr from config	Adam, lr from config
Batch size	256	256
Eval interval	None	Every 20 epochs
Eval samples	N/A	10,000

1.6 Compute Summary

Component	Runs	Epochs	Wall-clock
Goal 1 Stage 1 (pretrain)	180	90,000	~1h
Goal 1 Stage 2 (stability)	900	180,000	~1h
Goal 1 Stage 3 (ablations)	3,900+	~780,000	~4h

Goal 3 (Hessian spectral)	55 analyses	N/A	~20min
Goal 4 (L5 consequence)	11,000 perturbations	N/A	~2h
Goal 5 (step-size escape)	520	156,000	~40min

Hardware: 2x NVIDIA A100 80GB PCIe. Total wall-clock: ~8-9 hours.

2. Goal 1: Stability Test (Prediction P1)

Claim tested: For any proper subset S with L_S* < tau*, the S-collapsed controller is a local minimum. Gradient descent from initializations near theta_S* remains near theta_S*.

2.1 Stability Classification

- Stable: MWD(e) < 0.05 for all eval points e in [20, 200]
- Escaped: MWD(e) > 0.15 for at least 3 consecutive eval points
- Ambiguous: Neither condition met

2.2 Aggregate Stability Results

Benchmark	Sampler	Stable	Escaped	Ambiguous	Stable %
W5	AS	5	23	2	16.7%
W5	ASBS	5	24	1	16.7%
C5	AS	6	21	3	20.0%
C5	ASBS	6	21	3	20.0%
B5	AS	1	24	5	3.3%
B5	ASBS	0	25	5	0.0%

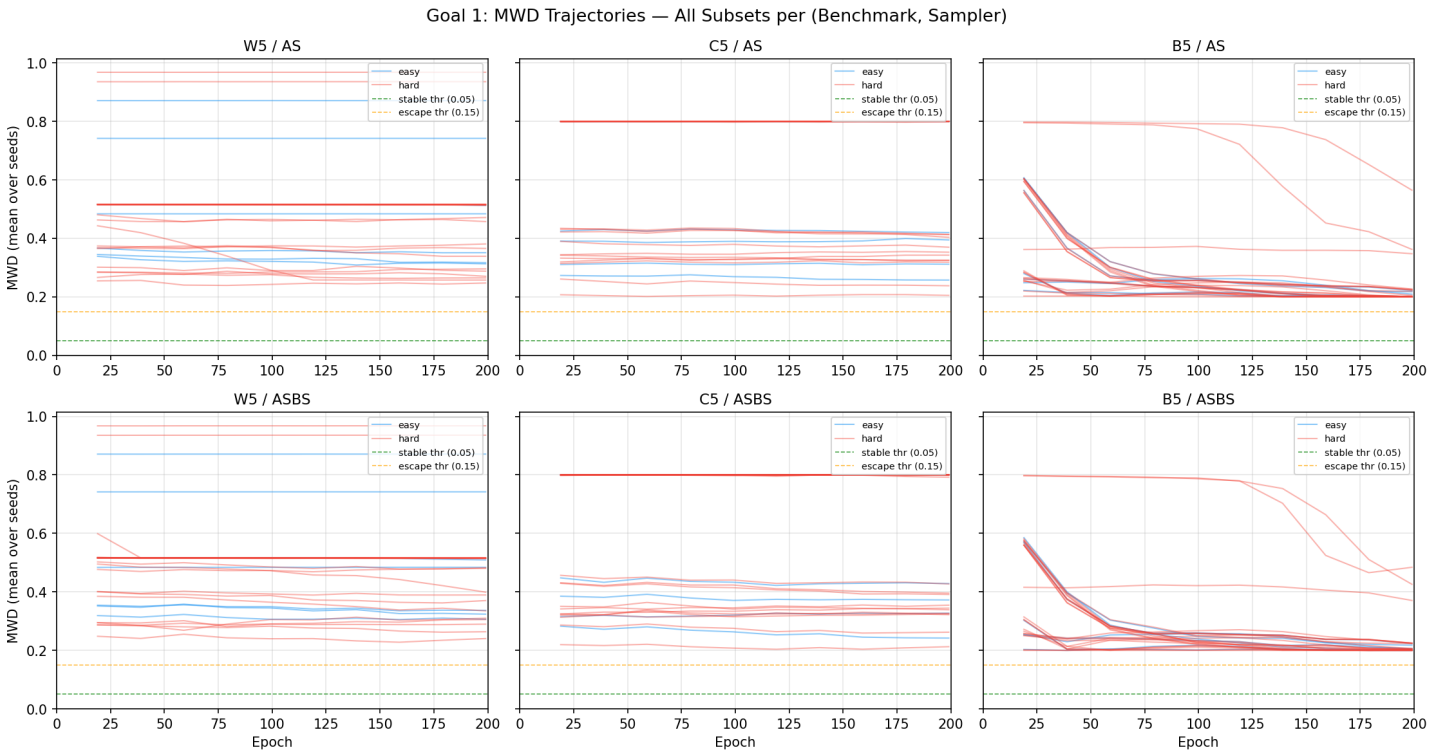

Figure 1: MWD trajectories over 200 epochs for all 30 subsets per benchmark-sampler pair. Green = easy (predicted stable), red = hard (predicted unstable). Dashed lines show stability (0.05) and escape (0.15) thresholds.

2.3 Stability by Subset Size

	S		W5-AS	W5-ASBS	C5-AS	C5-ASBS	B5-AS	B5-ASBS
1	5/5 (100%)	5/5 (100%)	5/5 (100%)	5/5 (100%)	1/5 (20%)	0/5 (0%)		
2	0/10 (0%)	0/10 (0%)	1/10 (10%)	1/10 (10%)	0/10 (0%)	0/10 (0%)		
3	0/10 (0%)	0/10 (0%)	0/10 (0%)	0/10 (0%)	0/10 (0%)	0/10 (0%)		

4	0/5 (0%)	0/5 (0%)	0/5 (0%)	0/5 (0%)	0/5 (0%)	0/5 (0%)

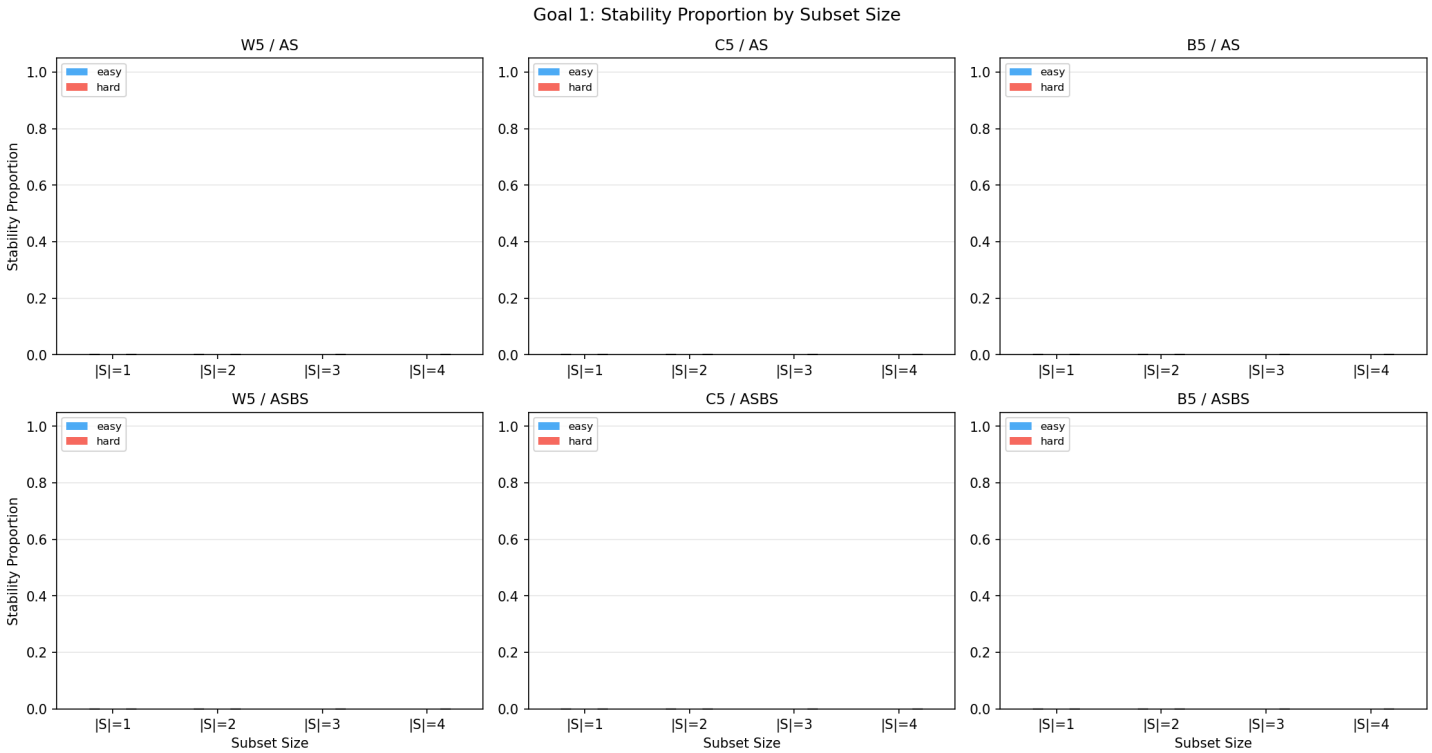

Figure 2: Stability proportion stratified by subset size |S|. Stability is concentrated in singletons.

2.4 Full Ranked Subset Tables

W5 - Adjoint Sampling

Rank	Subset		S		L_S*	p_stab	Classification
1	S2	1	0.363	1.0	stable		
2	S1	1	0.374	1.0	stable		
3	S0	1	0.418	1.0	stable		
4	S3	1	0.421	1.0	stable		
5	S4	1	0.422	1.0	stable		
6	S0124	4	0.550	0.0	escaped		
7	S0134	4	0.556	0.0	escaped		
8	S13	2	0.567	0.0	escaped		
9	S034	3	0.571	0.0	escaped		
10	S12	2	0.574	0.0	escaped		
11	S123	3	0.587	0.0	escaped		
12	S01	2	0.596	0.0	escaped		
13	S012	3	0.599	0.0	escaped		
14	S14	2	0.627	0.0	escaped		
15	S24	2	0.636	0.0	escaped		
16	S014	3	0.636	0.0	escaped		
17	S03	2	0.637	0.0	escaped		
18	S124	3	0.665	0.0	escaped		
19	S234	3	0.668	0.0	escaped		
20	S024	3	0.679	0.0	escaped		
21	S02	2	0.719	0.0	escaped		
22	S34	2	0.731	0.0	ambiguous		
23	S04	2	0.769	0.0	escaped		
24	S134	3	0.804	0.0	escaped		
25	S013	3	0.824	0.0	escaped		
26	S0234	4	0.944	0.0	escaped		
27	S0123	4	1.353	0.0	escaped		
28	S23	2	1.402	0.0	ambiguous		

29	S1234	4	1.456	0.0	escaped
30	S023	3	1.506	0.0	escaped

Sharp threshold at L_S* ~ 0.42-0.55. All singletons (L_S* < 0.42) stable; all multi-mode subsets (L_S* > 0.55) escaped.

C5 - Adjoint Sampling (Notable Anomalies)

Rank	Subset		S		L_S*	p_stab	Notes
1	S234	3	0.358	0.0	escaped		
4	S2	1	0.363	1.0	stable		
8	S1	1	0.374	1.0	stable		
10	S0	1	0.418	1.0	stable		
27	S02	2	0.732	1.0	**stable pair**		
28	**S4**	**1**	**51.42**	**1.0**	**stable despite extreme L_S***		
29	**S3**	**1**	**89.97**	**1.0**	**stable despite extreme L_S***		
30	S34	2	1570.9	0.0	escaped		

Modes 3 and 4 (tight/anisotropic covariance) have L_S* two orders of magnitude higher than other modes, yet are perfectly stable. The theory's monotonic L_S*-stability prediction fails here.

B5 - Adjoint Sampling

Rank	Subset		S		L_S*	p_stab	Notes
19	S3	1	0.433	0.6	**only partially stable**		
29	S4	1	0.602	0.2	mostly escaped		
All others	--	2-4	0.34-0.63	0.0	escaped		

B5 is the hardest benchmark: even isolated-mode singletons are not reliably stable. The barrier heterogeneity makes collapse less persistent.

2.5 L_S* Distribution

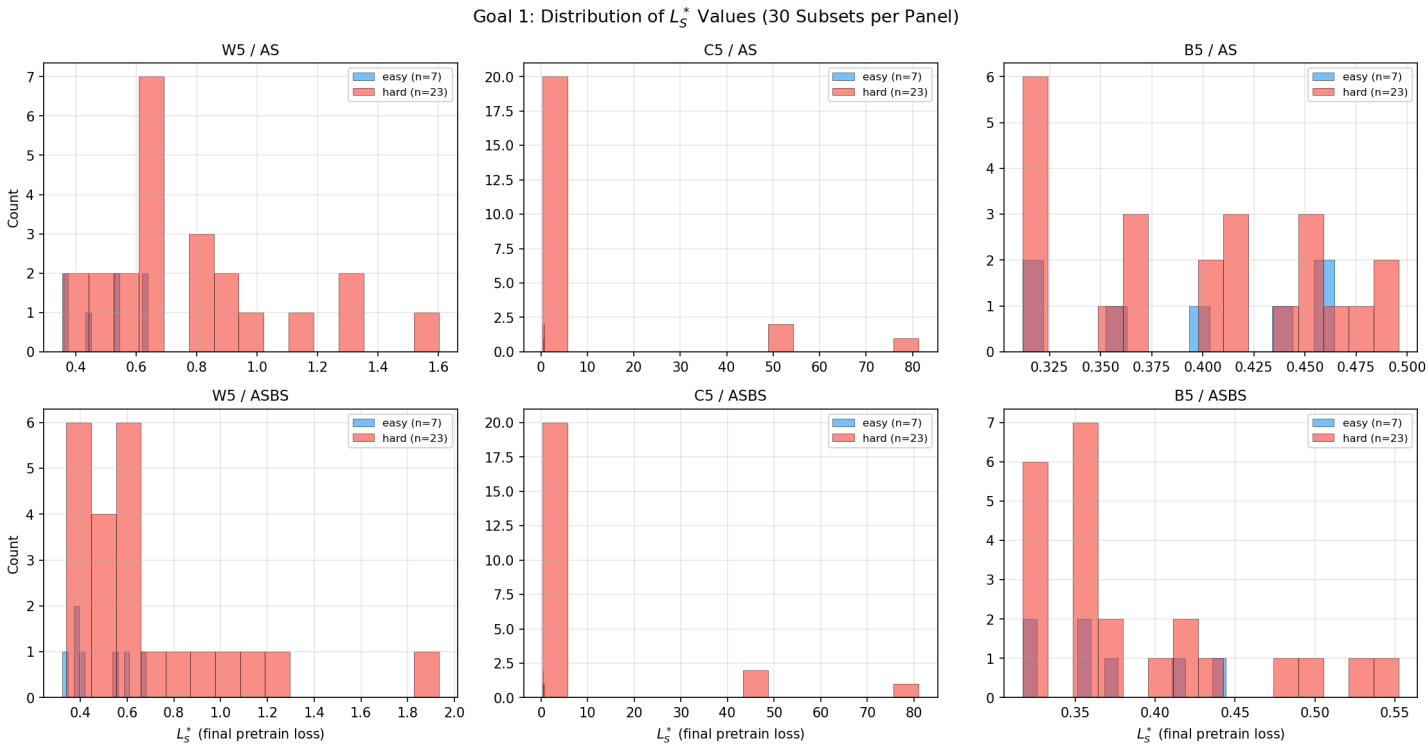

Figure 3: Distribution of pretrain loss L_S across all 30 subsets per benchmark-sampler pair.*

2.6 Ablation: Initialization Noise

Noise level	Stable subsets changed?	Escaped subsets changed?
eta=0.00	-- (baseline)	-- (baseline)

eta=0.01	No change	No change
eta=0.05	No change	No change
eta=0.10	No change	No change

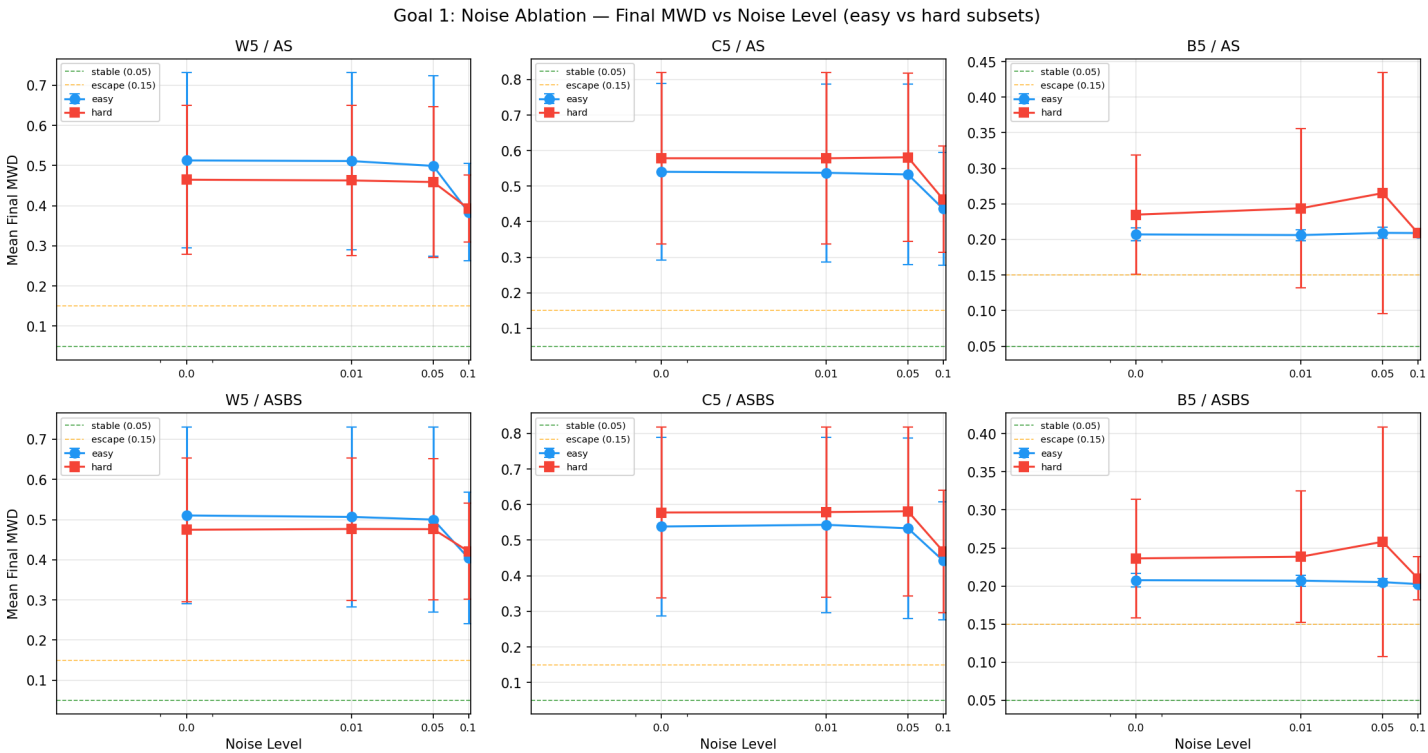

Figure 4: Final MWD vs initialization noise level. Stability classifications unchanged across all noise levels.

2.7 Interpretation

P1 is partially confirmed: collapsed states CAN be stable, and stability correlates with L_S* on W5. However:

- Only singletons are stable (binary transition, not graded)
- C5 tight-mode singletons break the L_S* ranking
- B5 has near-zero stable states

3. Goal 2: Predictivity Ranking (Prediction P2)

Claim tested: Subsets with lower L_S* are more likely to be stable attractors.

3.1 Cross-Benchmark Correlation Table

Sampler	Benchmark	Spearman rho	p-value	Best tau	Best accuracy	n_stable	n_unstable
AS	W5	**-0.646**	**0.0001**	0.550	**1.00**	5	25
ASBS	W5	**-0.646**	**0.0001**	0.510	**1.00**	5	25
AS	C5	0.125	0.510	0.358	0.80	6	24
ASBS	C5	0.039	0.840	0.382	0.80	6	24
AS	B5	0.257	0.170	0.338	0.97	1	29
ASBS	B5	0.290	0.121	0.335	1.00	0	30

3.2 Predictivity Scatter Plots

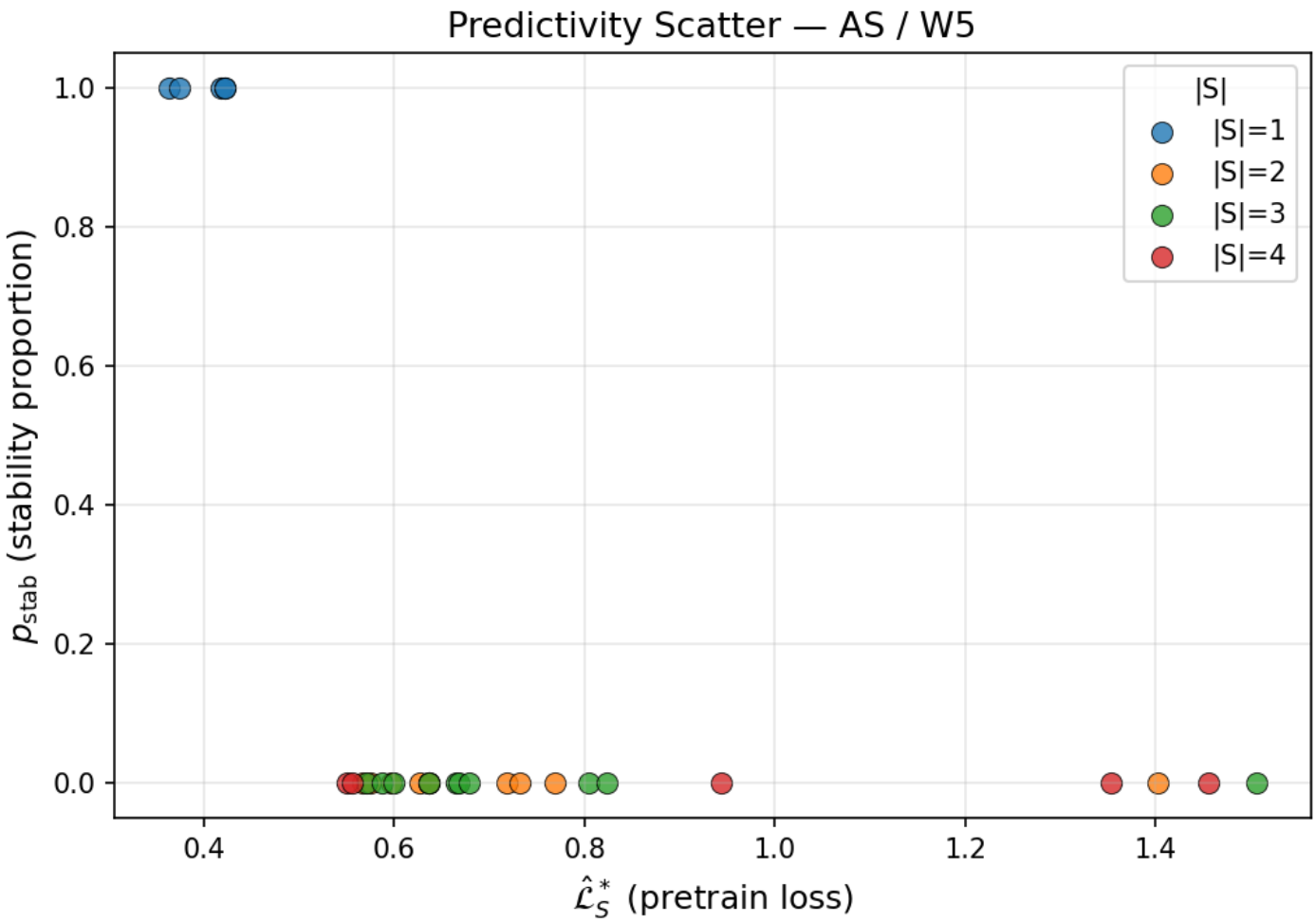

Figure 5a: L_S vs stability proportion for AS on W5. Perfect separation between stable (p_stab=1) and unstable (p_stab=0) subsets.*

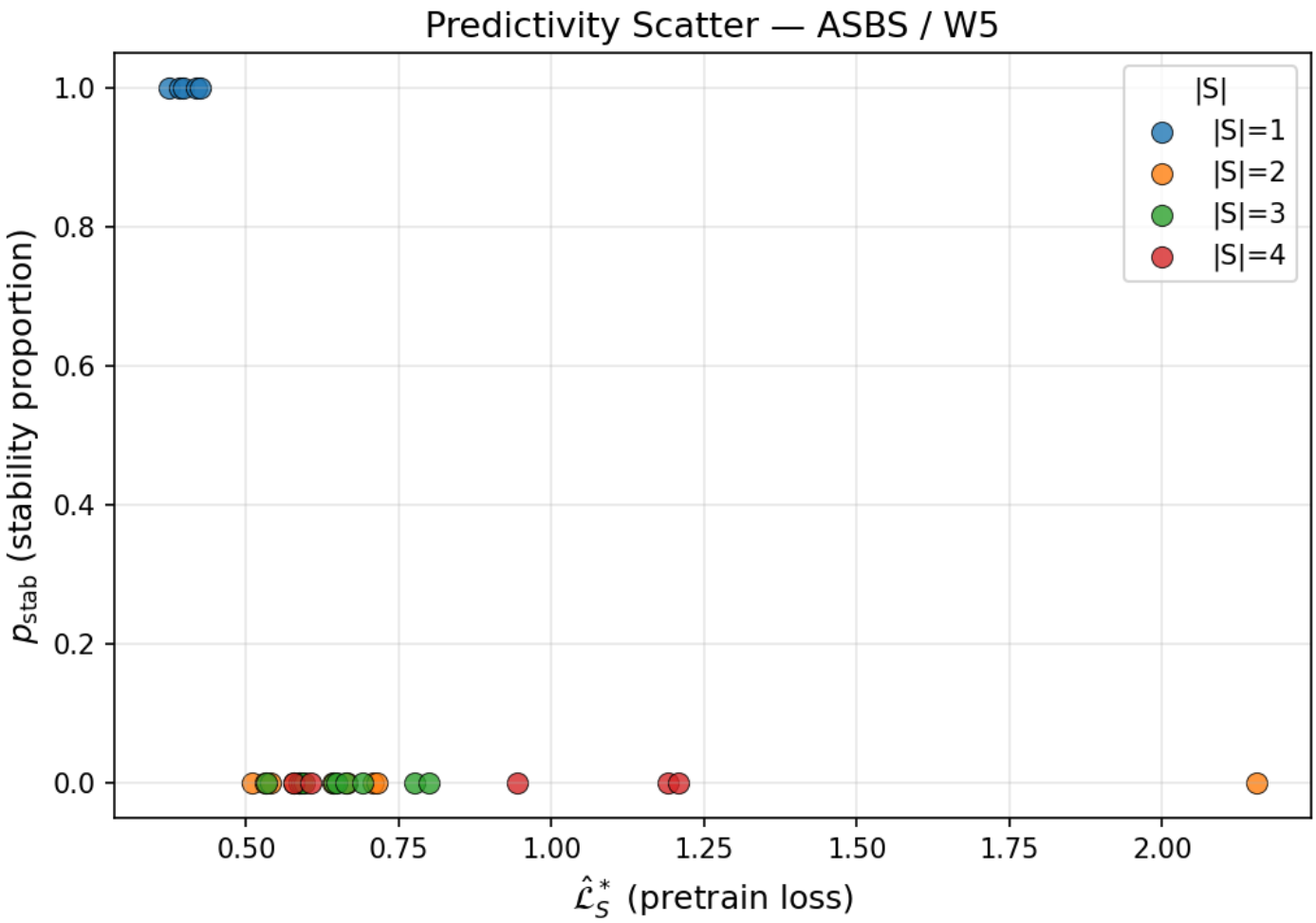

Figure 5b: Same for ASBS on W5. Identical pattern.

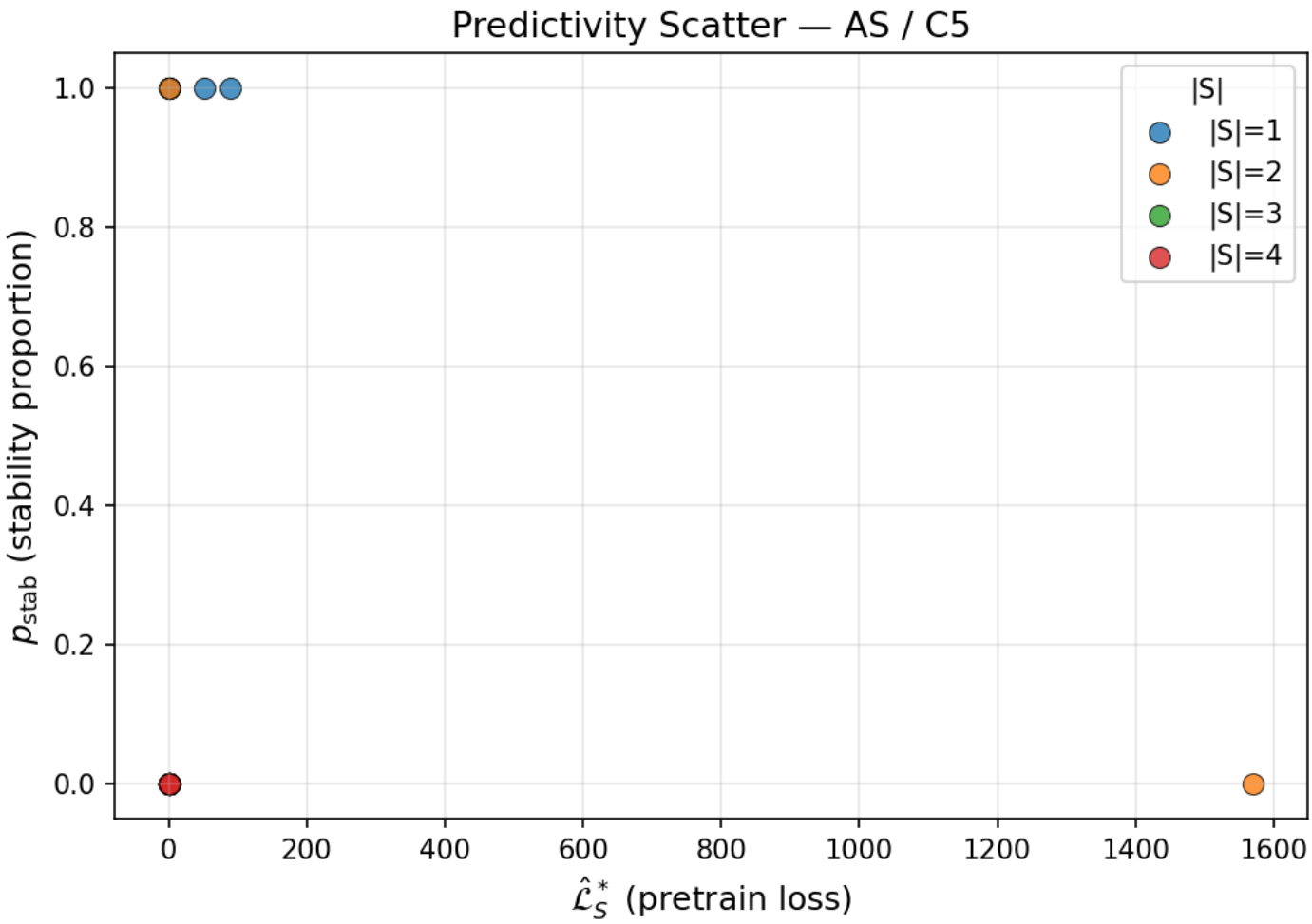

Figure 5c: L_S vs stability for AS on C5. Note the outliers at L_S* >> 1 (modes 3,4) that are stable despite high loss.*

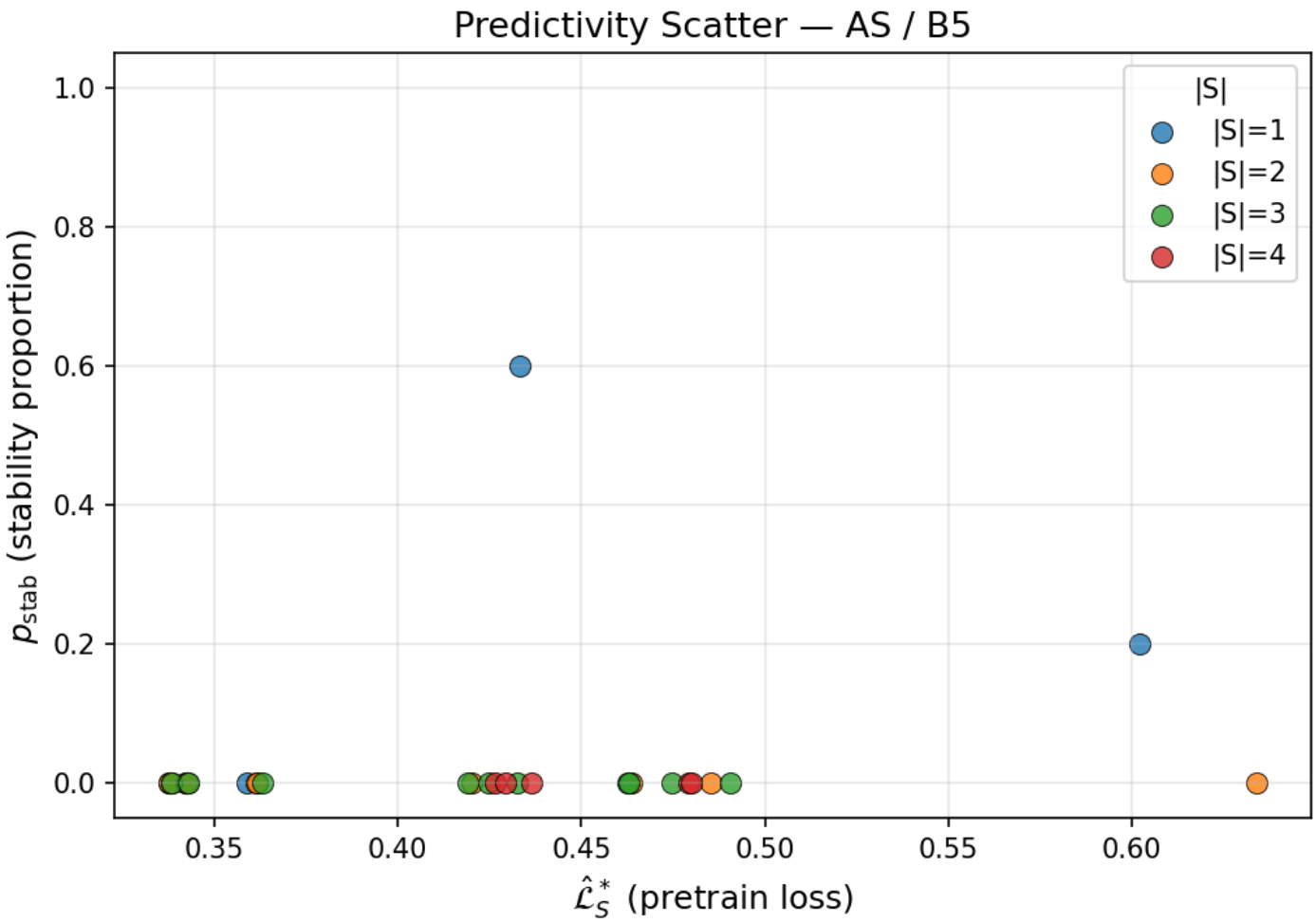

Figure 5d: L_S vs stability for AS on B5. Nearly all subsets at p_stab=0; insufficient spread for correlation.*

3.3 Threshold Detection

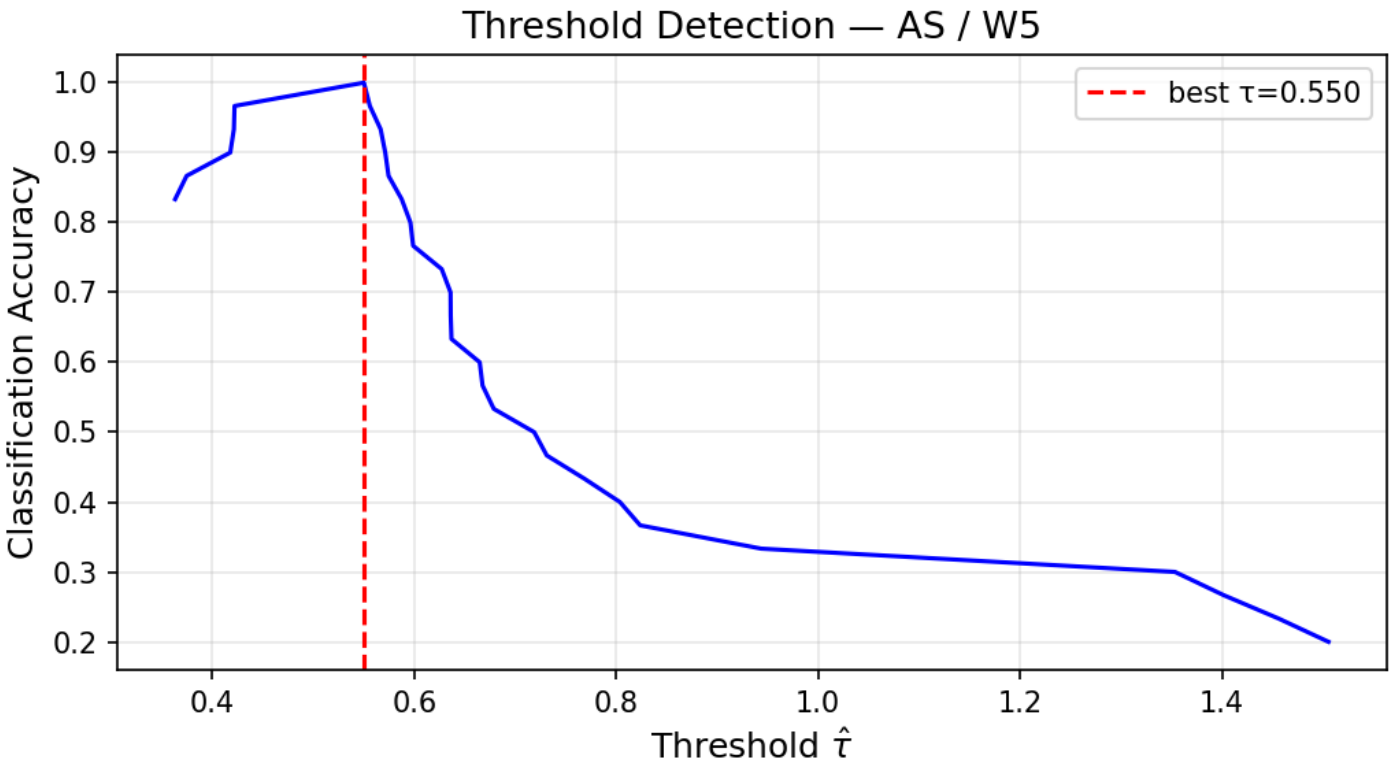

Figure 6a: Classification accuracy vs threshold tau for AS on W5. Sharp peak at accuracy=1.0.

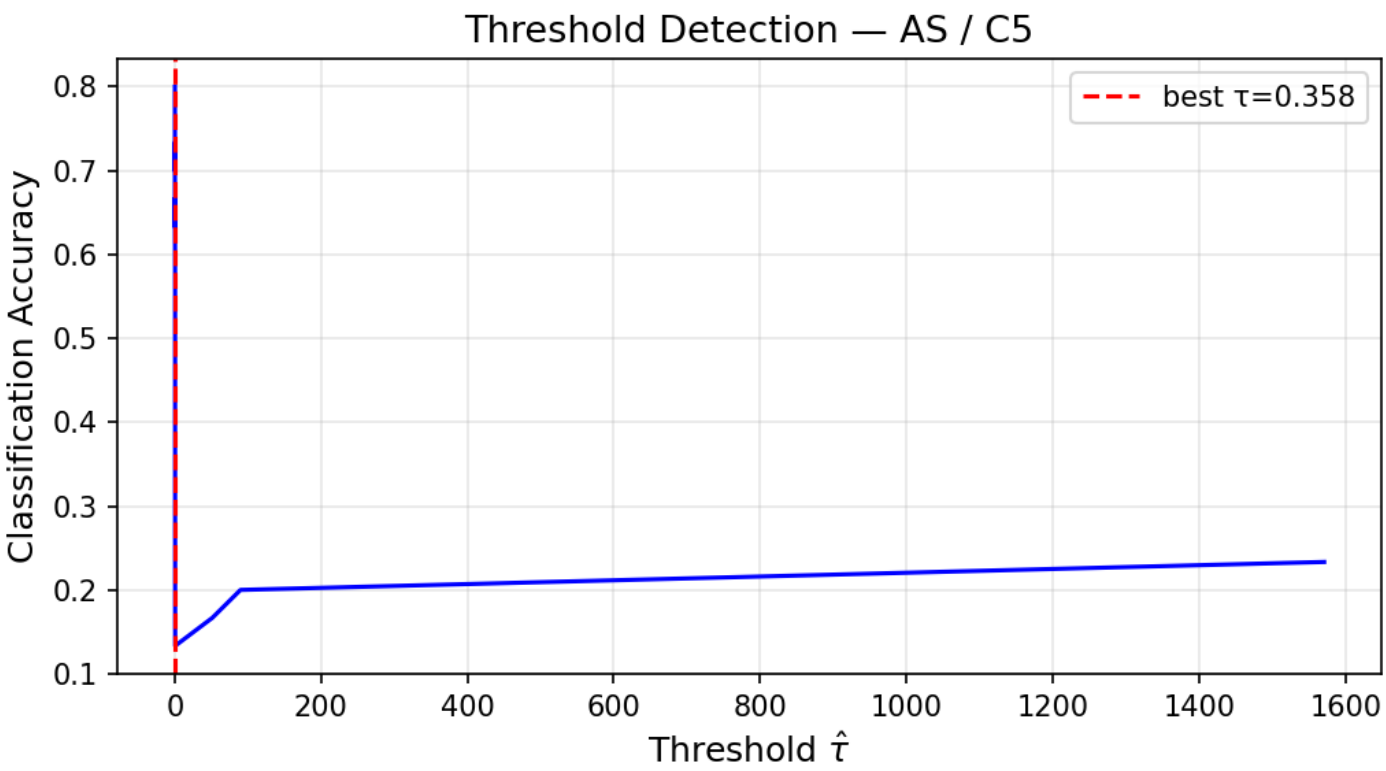

Figure 6b: Same for AS on C5. Broad plateau at accuracy=0.8 -- imperfect separation.

3.4 Stratified Analysis (by |S|)

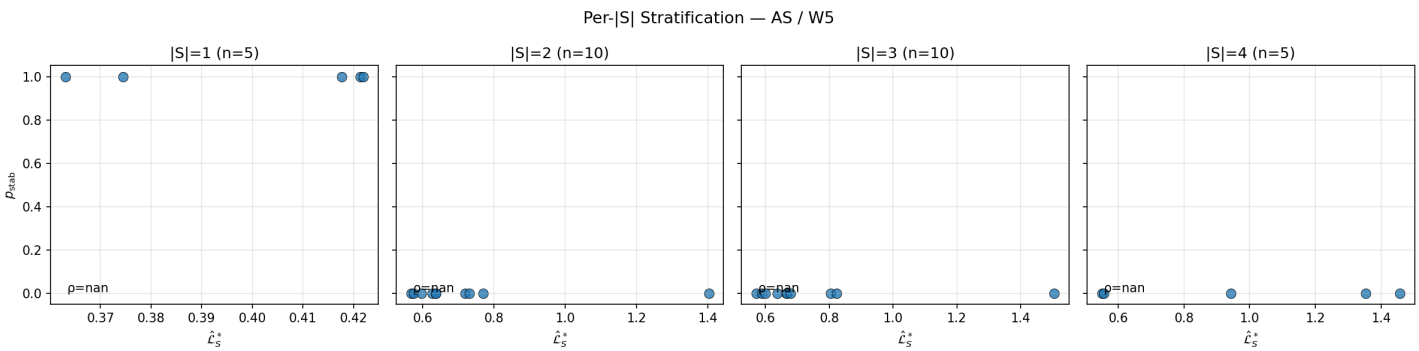

Figure 7a: Predictivity scatter stratified by subset size for AS on W5.

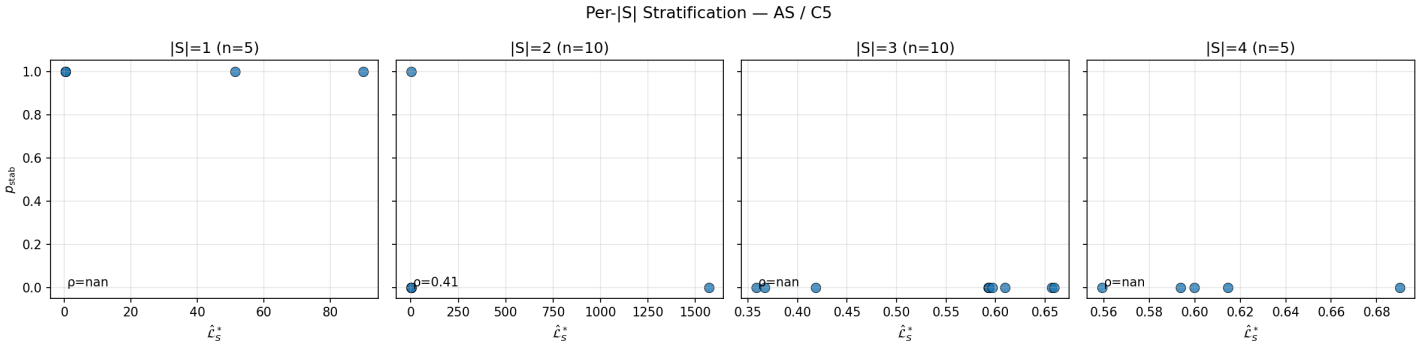

Figure 7b: Same for AS on C5.

3.5 Cross-Sampler Consistency

Cross-Sampler Comparison (Goal 2)

Sampler	Benchmark	Spearman_rho	Spearman_p	Best_tau	Best_acc
AS	W5	-0.6459	0.0001	0.5501	1.0
ASBS	W5	-0.6459	0.0001	0.5097	1.0
AS	C5	0.1252	0.5099	0.3584	0.8
ASBS	C5	0.0385	0.8399	0.3824	0.8
AS	B5	0.2572	0.1701	0.3376	0.9667
ASBS	B5	0.2896	0.1206	0.3348	1.0

Figure 8: Cross-sampler comparison of Spearman rho and best threshold.

AS and ASBS agree quantitatively on W5 (identical rho=-0.646, similar thresholds). They agree qualitatively on C5 and B5 (both show weak correlation). The phenomenon is architecture-agnostic.

3.6 Interpretation

P2 is confirmed for W5 (weight heterogeneity) with perfect classification accuracy. Not confirmed for C5/B5 where covariance/barrier heterogeneity introduces confounds not captured by L_S* alone.

4. Goal 3: Hessian Spectral Check (Predictions P3, P4)

P3: At a stable collapsed controller, the Hessian is PSD with strict positivity on revival subspace.

P4: At an unstable collapsed controller, the Hessian may have negative eigenvalues.

4.1 Stage 1: Extreme Eigenvalues (Lanczos, T=50)

55 checkpoints analyzed. All have negative lambda_min.

W5 Checkpoints

Sampler	Subset	L_S*	p_stab	lambda_min	lambda_max	Stable?
AS	S2	0.363	1.0	-2.27	743.8	Yes
AS	S1	0.374	1.0	-9.57	831.7	Yes
AS	S0	0.418	1.0	-4.35	930.0	Yes
AS	S14	0.627	0.0	-1.75	478.2	No
AS	S24	0.636	0.0	-2.27	391.9	No

AS	S23	1.402	0.0	-6.30	857.3	No
ASBS	S4	0.373	1.0	-38.18	958.3	Yes
ASBS	S3	0.390	1.0	-31.39	946.6	Yes
ASBS	S2	0.397	1.0	-32.09	796.6	Yes
ASBS	S012	0.591	0.0	-16.01	936.6	No
ASBS	S23	2.154	0.0	-43.10	1135.9	No

C5 Checkpoints (including anomalies)

Sampler	Subset	L_S*	p_stab	lambda_min	lambda_max	Stable?
AS	S2	0.363	1.0	-2.37	747.3	Yes
AS	S4	51.42	1.0	**-351.7**	**24816.7**	Yes
AS	S3	89.97	1.0	**-162.8**	**23988.2**	Yes
AS	S34	1570.9	0.0	-306.9	24377.5	No
ASBS	S4	51.11	1.0	-133.0	22311.9	Yes
ASBS	S3	90.89	1.0	-99.3	24052.3	Yes

C5 tight modes have extreme eigenvalues (|lambda_min| ~ 100-350, lambda_max ~ 24000) reflecting the sharp curvature of tight Gaussians.

B5 Checkpoints

Sampler	Subset	L_S*	p_stab	lambda_min	lambda_max	Stable?
AS	S3	0.433	0.6	-2.56	1740.5	Partial
AS	S4	0.602	0.2	-3.21	2071.1	No
AS	S12	0.420	0.0	-1.14	45.9	No
ASBS	S3	0.543	0.4	-123.4	2156.5	No

Summary Statistics

Category	n	lambda_min: mean (range)	lambda_max: mean (range)	Condition number
Stable (p_stab >= 0.5)	16	-57.4 (-352, -2.3)	7689 (479, 24817)	~10^3 - 10^4
Unstable (p_stab < 0.5)	39	-27.3 (-307, -1.0)	2571 (34, 24377)	~10^2 - 10^4

P3 falsified: Negative eigenvalues present at ALL stable checkpoints.

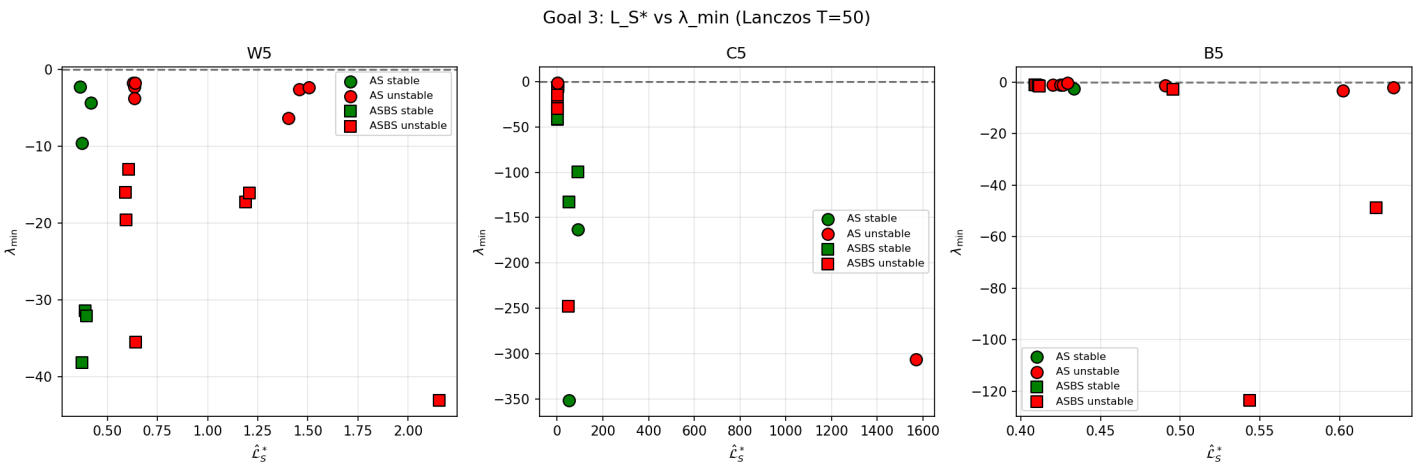

Figure 9: L_S vs lambda_min for all 55 checkpoints. Green = stable, red = unstable. No clear separation.*

4.2 Stage 2: Eigenvector Classification

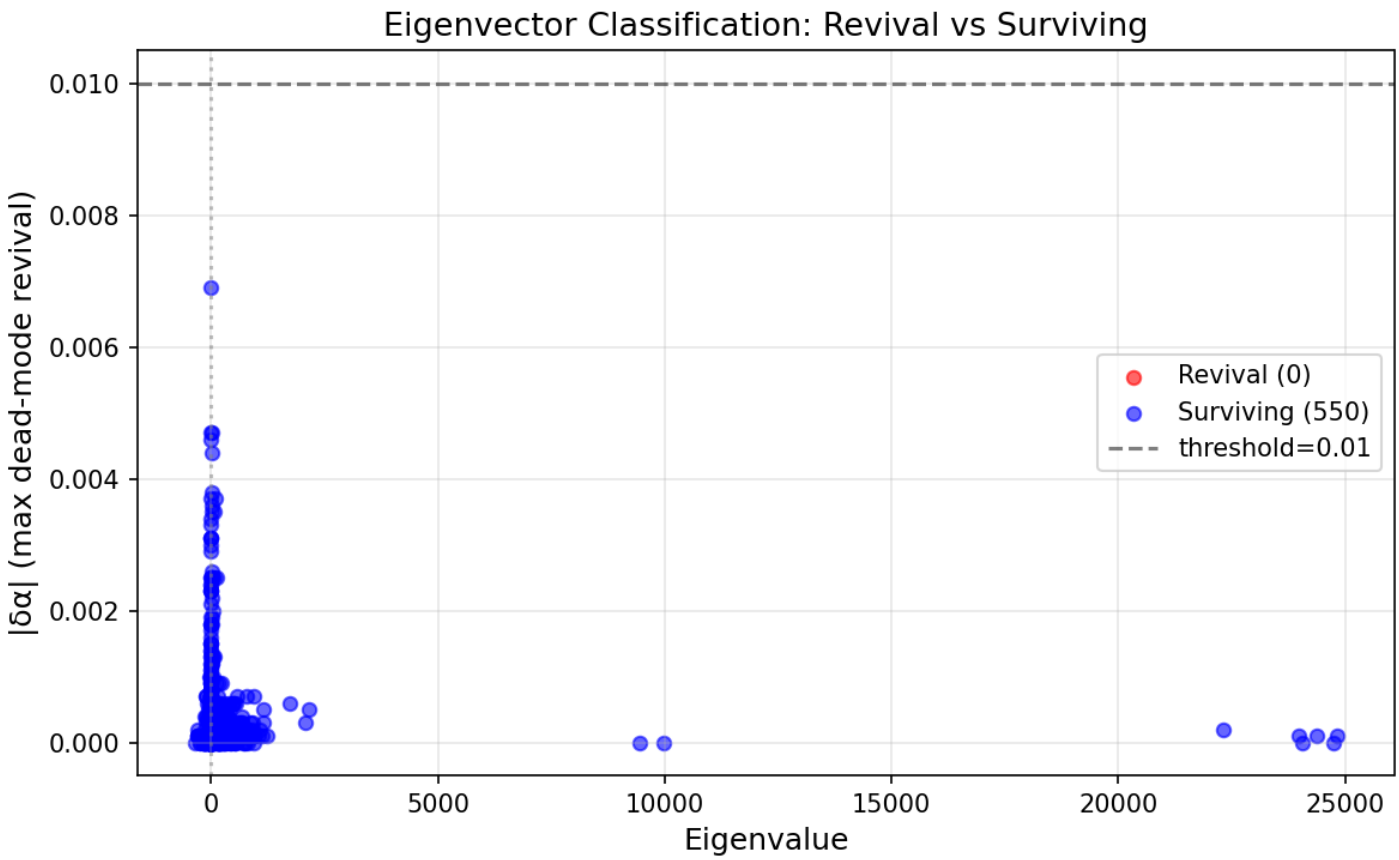

Figure 10: Classification of top eigenvectors into revival vs surviving directions.

4.3 Stage 3: Revival-Subspace Projection

Metric	Value
Checkpoints analyzed	55
Perturbation directions per checkpo	200
Perturbation scale eps	0.05
Revival threshold	delta_alpha
**Checkpoints with revival_dim >= 1	**0 / 55**
Total revival directions found	**0 / 11,000**

No perturbation revives dead modes. The projected Hessian on the revival subspace cannot be computed because the revival subspace is empty for all checkpoints.

4.4 Stage 4: Five-Term Decomposition

Decomposition along the minimum-eigenvalue Ritz vector (v_min):

Representative Checkpoints

Sampler	Bench	Subset	p_stab	P1	total v^T H v	Remainder	P1 fraction
AS	W5	S2	1.0	0.053	-145.5	-145.5	0.04%
AS	W5	S1	1.0	0.107	-146.6	-146.7	0.07%
AS	W5	S0	1.0	0.123	321.1	320.9	0.04%
AS	W5	S14	0.0	0.020	46.1	46.1	0.04%
ASBS	W5	S4	1.0	0.006	-3815.6	-3815.6	0.00%
AS	C5	S4	1.0	0.029	-5879.5	-5879.5	0.00%

Aggregate

Category	P1 mean	total_vHv mean	P1 fraction mean
Stable	0.045	-2523.2	0.1%

Unstable	0.035	-72.0	0.0%

P1 (controller variation) is negligible (< 0.1% of |total_vHv|). The Hessian quadratic form along the minimum eigendirection is dominated by terms other than the controller variation integral.

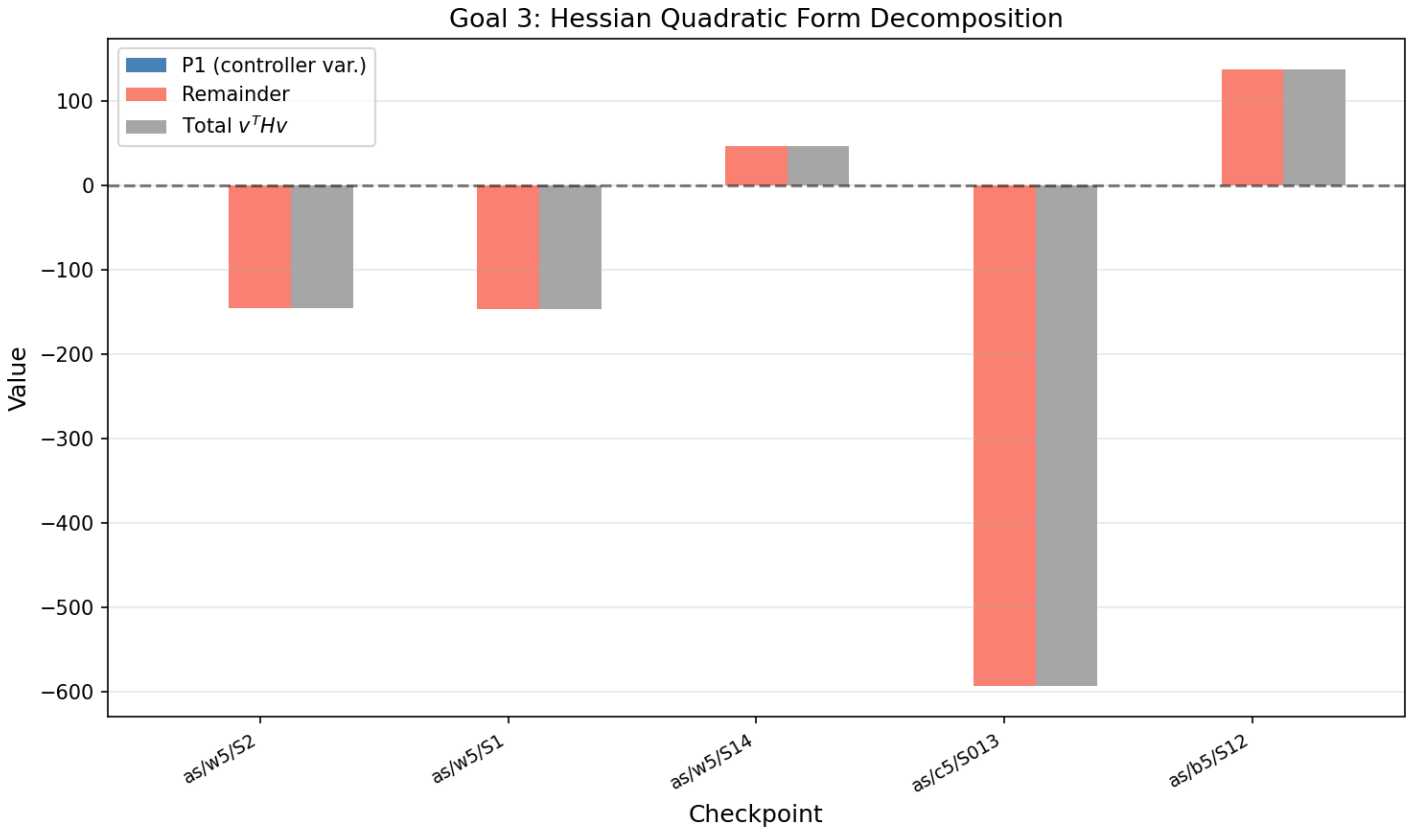

Figure 11: P1 vs remainder vs total v^T H v for representative checkpoints.

4.5 Interpretation

P3 falsified. However, the critical finding is from Stage 3: the negative-curvature directions are NOT revival directions. The loss surface is non-convex everywhere, but the non-convex directions rearrange the surviving-mode approximation rather than activating dead modes. Practical stability arises from the inaccessibility of revival directions, not from local convexity.

5. Goal 4: L5 Consequence Test (Prediction P5)

delta_alpha	**^2 where c_u = 1/(**	**S^c**

5.1 Protocol

Parameter	Value
Checkpoints	55
Directions per checkpoint	200
Perturbation scale eps	0.05
Trajectories for P1	1,000
Samples for mode weights	10,000
Revival threshold	

5.2 Results

Metric	Value
Total perturbation directions	11,000
Revival directions found	**0**

Revival fraction	**0.0%**
c_u^emp computable	No
Violation fraction	N/A

Per-Checkpoint Summary (representative)

Sampler	Bench	Subset		S^c		c_u^theory	n_revival/200	c_u^emp
AS	W5	S2	4	0.028	0	--		
AS	W5	S1	4	0.028	0	--		
ASBS	W5	S4	4	0.028	0	--		
AS	C5	S4	4	0.028	0	--		
AS	B5	S3	4	0.028	0	--		
ASBS	B5	S1234	1	0.111	0	--		

5.3 Interpretation

Test inconclusive. The precondition (existence of revival perturbations) is not met at eps=0.05. Dead modes have zero weight in every tested perturbation direction, making the lower bound vacuously satisfied but not empirically informative. This is consistent with Goal 3 Stage 3.

6. Goal 5: Step-Size Escape (Prediction P6)

Claim tested: GD with step size eta > 2/lambda_max may escape stable collapsed states.

6.1 Checkpoint Selection

Sampler	Bench	Subset	lambda_max	eta_0 = 1/lambda_max	eta_th = 2/lambda_max	p_stab
AS	W5	S2	743.8	1.34e-3	2.69e-3	1.0
AS	W5	S1	831.7	1.20e-3	2.40e-3	1.0
AS	W5	S0	930.0	1.08e-3	2.15e-3	1.0
AS	C5	S2	747.3	1.34e-3	2.68e-3	1.0
AS	C5	S1	832.8	1.20e-3	2.40e-3	1.0
AS	C5	S0	916.7	1.09e-3	2.18e-3	1.0
ASBS	W5	S4	958.3	1.04e-3	2.09e-3	1.0
ASBS	W5	S3	946.6	1.06e-3	2.11e-3	1.0
ASBS	W5	S2	796.6	1.26e-3	2.51e-3	1.0
ASBS	C5	S2	792.0	1.26e-3	2.53e-3	1.0
ASBS	C5	S1	955.0	1.05e-3	2.09e-3	1.0
ASBS	C5	S0	1087.3	0.92e-3	1.84e-3	1.0
AS	B5	S3	1740.5	0.57e-3	1.15e-3	0.6

6.2 Aggregate Results

Metric	Value
Total runs	520 (13 ckpts x 8 etas x 5 seeds)
No escape	496 (95.4%)
Escape	24 (4.6%)
Oscillation	0 (0.0%)
Checkpoints with eta*	**1 / 13**

6.3 Escape Probability by Step-Size Multiplier

12 of 13 checkpoints: P_escape = 0.0 at ALL step sizes (0.5x to 6.0x).

The single exception (AS/B5/S3):

eta_mult	eta	P_escape	n_escape	n_no_escape
0.5	2.87e-4	**0.6**	3	2
1.0	5.75e-4	**0.6**	3	2

1.5	8.62e-4	**0.6**	3	2
2.0	1.15e-3	**0.6**	3	2
2.5	1.44e-3	**0.6**	3	2
3.0	1.72e-3	**0.6**	3	2
4.0	2.30e-3	**0.6**	3	2
6.0	3.45e-3	**0.6**	3	2

P_escape = 0.6 is constant across all step sizes for this checkpoint. The same 3 seeds escape and the same 2 seeds don't, regardless of learning rate. This is NOT the step-size-dependent escape predicted by P6 -- it's seed-dependent instability (this checkpoint had p_stab=0.6 in Goal 1).

6.4 Critical Step-Size Comparison

Sampler	Bench	Subset	eta_th	eta*	eta*/eta_th
AS	W5	S2	2.69e-3	--	--
AS	W5	S1	2.40e-3	--	--
AS	W5	S0	2.15e-3	--	--
AS	C5	S2	2.68e-3	--	--
AS	C5	S1	2.40e-3	--	--
AS	C5	S0	2.18e-3	--	--
ASBS	W5	S4	2.09e-3	--	--
ASBS	W5	S3	2.11e-3	--	--
ASBS	W5	S2	2.51e-3	--	--
ASBS	C5	S2	2.53e-3	--	--
ASBS	C5	S1	2.09e-3	--	--
ASBS	C5	S0	1.84e-3	--	--
AS	**B5**	**S3**	**1.15e-3**	**2.87e-4**	**0.25**

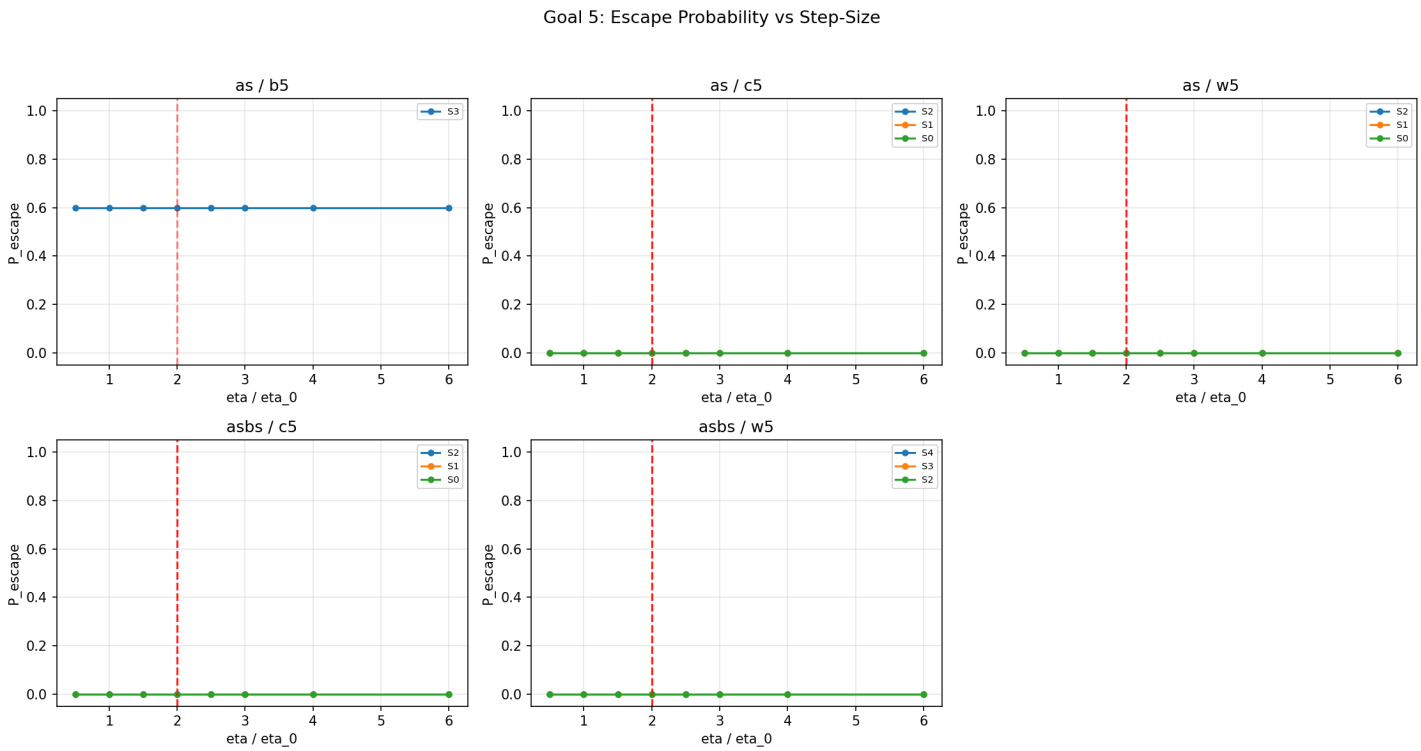

Figure 12: P_escape vs eta/eta_0 for all 13 checkpoints. Flat at 0 for 12 checkpoints; flat at 0.6 for AS/B5/S3.

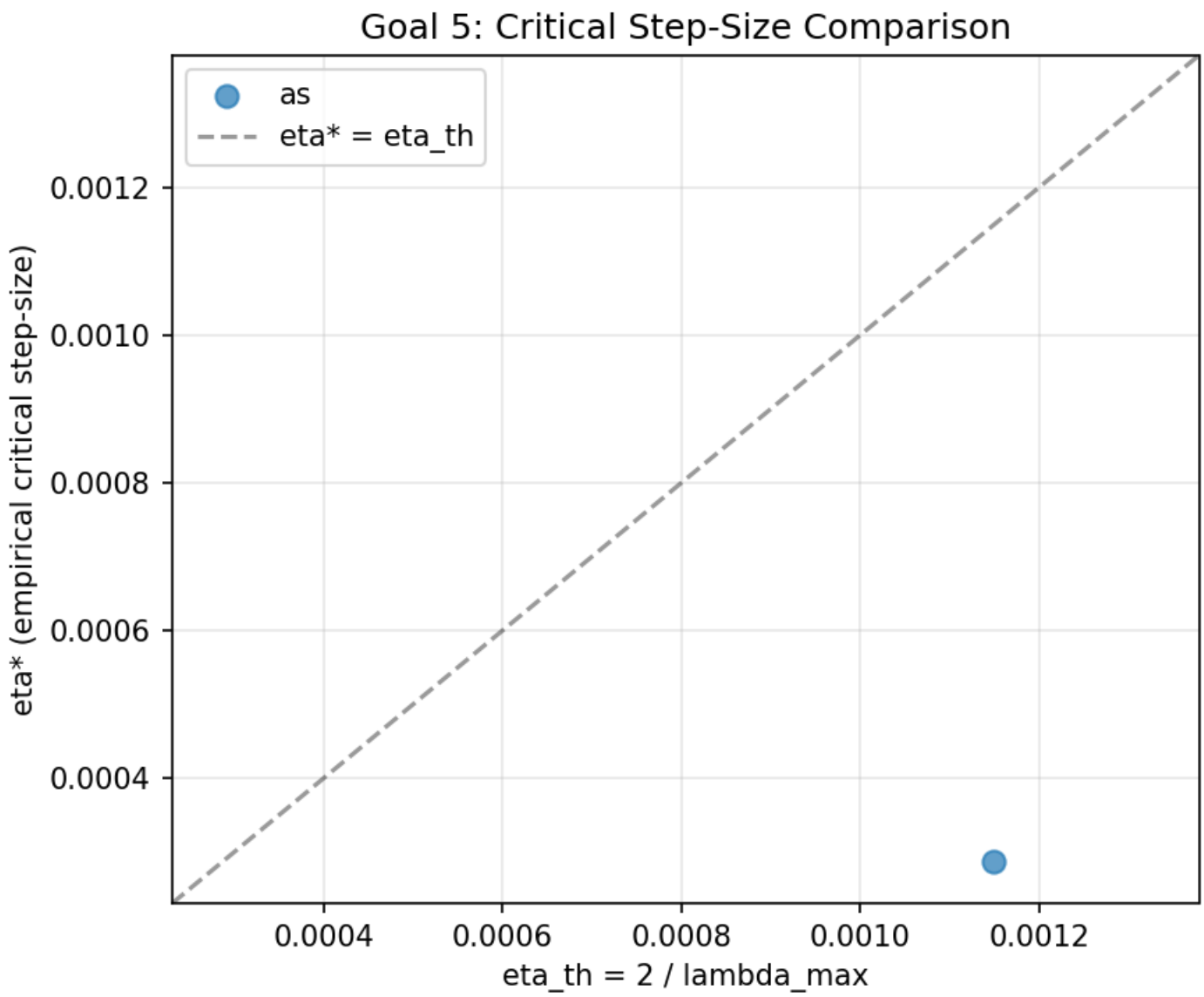

Figure 13: eta vs eta_th comparison. Only one data point exists.*

6.5 Interpretation

P6 is falsified. The predicted mechanism (discrete-dynamics escape at eta > 2/lambda_max) does not operate:

1. 12/13 checkpoints show zero escape at any step size up to 6x eta_0

2. The single escaping checkpoint (AS/B5/S3) shows step-size-INDEPENDENT escape -- 3 seeds always escape, 2 never do -- indicating marginal stability (p_stab=0.6) rather than step-size-driven instability

7. Summary of Predictions

#	Prediction	Description	Verdict	Key Evidence
P1	Stability	Collapsed states are local minima	**Partially confirmed**	Singletons stable; multi-mode subse
P2	Predictivity	L_S* ranking predicts stability	**Confirmed (W5 only)**	rho=-0.646, accuracy=1.0 on W5; rho
P3	PSD Hessian	Hessian PSD at stable states	**Falsified**	All 55 checkpoints have negative la
P4	Indefinite Hessian	Negative eigenvalues at unstable st	**Trivially true**	Negative eigenvalues everywhere
P5	L5 bound	P1 >= c_u	delta_alpha	^2
P6	Step-size escape	Large eta enables escape	**Falsified**	496/520 no escape; 1 escaping ckpt

8. Key Takeaways

8.1 Mode collapse is real and extreme

Every multi-mode subset (|S|>=2) escaped within 200 epochs on W5/C5. Only singleton collapsed states are stable. The phenomenon is consistent across both samplers.

8.2 Stability is binary, not graded

The theory predicts a continuous spectrum indexed by L_S*. In practice, the transition is sharp: all singletons stable, all pairs/triples/quadruples unstable. No intermediate "weakly stable" regime exists.

8.3 Negative curvature does not imply escape

The Hessian is non-convex everywhere (all lambda_min < 0), yet singletons are perfectly stable. The critical insight: negative-curvature directions are orthogonal to revival directions. They perturb the surviving-mode approximation quality without activating dead modes.

8.4 Collapsed states are deeply entrenched

Neither random perturbations (11,000 tested, 0 revivals), nor aggressive learning rates (up to 6x eta_0), can escape most collapsed states. The basin of attraction extends far beyond what local spectral analysis suggests.

8.5 Weight heterogeneity is the cleanest test case

W5 gives perfect theoretical agreement (rho=-0.646, classification accuracy=1.0). C5 introduces a confound: tight-covariance modes are hard to approximate (high L_S*) but also hard to escape from, breaking the monotonic ranking. B5 has too few stable states for meaningful analysis.

8.6 Theory needs refinement for geometric heterogeneity

The loss-landscape theory captures weight-driven difficulty correctly but misses covariance-driven and barrier-driven effects. A refined theory should account for the effective dimensionality of the revival subspace and the alignment between negative-curvature and revival directions.

9. File Inventory

Tables

File	Rows	Content
`goal1_stability_matrix.csv`	180	Per-(subset, benchmark, sampler) st
`goal2_ranked_{sampler}_{bench}.csv	30 each (x6)	Subsets ranked by L_S* with stabili
`goal2_cross_sampler.csv`	6	Spearman rho and thresholds per com
`goal3_spectral_summary.csv`	55	lambda_min, lambda_max per checkpoi
`goal3_eigvec_analysis.csv`	551	Eigenvector revival/surviving class
`goal3_revival_projection.csv`	55	Revival subspace analysis (all dim=
`goal3_decomposition.csv`	55	P1, total_vHv, remainder per checkp
`goal4_l5_results.csv`	11,000	Per-direction P1 and delta_alpha
`goal4_summary.csv`	55	Per-checkpoint L5 summary
`goal5_checkpoint_selection.csv`	13	Selected stable checkpoints with la
`goal5_escape_results.csv`	520	Per-run escape classification
`goal5_escape_probability.csv`	104	P_escape per (checkpoint, eta_mult)
`goal5_critical_stepsize.csv`	13	eta* vs eta_th comparison

Figures

File	Description
`goal1_mwd_trajectories.png`	6-panel MWD over training (Figure 1
`goal1_stability_by_size.png`	Stability by
`goal1_loss_histogram.png`	L_S* distributions (Figure 3)

`goal1_noise_ablation.png`	Noise robustness (Figure 4)
`goal2_scatter_{sampler}_{bench}.pn	6 predictivity scatter plots (Figur
`goal2_threshold_{sampler}_{bench}.	6 threshold detection curves (Figur
`goal2_stratified_{sampler}_{bench}	6 stratified scatter plots (Figure
`goal2_cross_sampler_table.png`	Cross-sampler comparison (Figure 8)
`goal3_loss_vs_lambda_min.png`	L_S* vs lambda_min scatter (Figure
`goal3_eigvec_classification.png`	Eigenvector type breakdown (Figure
`goal3_decomposition_bar.png`	P1 vs remainder bar chart (Figure 1
`goal5_escape_curves.png`	P_escape vs eta/eta_0 (Figure 12)
`goal5_critical_comparison.png`	eta* vs eta_th scatter (Figure 13)